

WHODUNNIT?

SKILL BUILDERS



**Up-and-moving
skill practice
activity!**

**Mean, Median,
Mode & Range**



Whodunnit?

Mean, Median, Mode and Range

Ideal Unit:	Measures of Center	Time Range:	20-40 Minutes	Supplies:	Pencil & Paper
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Common Core Alignment:

This particular activity was mapped to these common core standards. *Whodunnit* activities are ideal as a small group skill practice.

6.SP.A.3	Recognize that a measure of center for a numerical data set summarizes all of its values with a single number, while a measure of variation describes how its values vary with a single number.
6.SP.B.5	Summarize numerical data sets in relation to their context, such as by: Giving quantitative measures of center (median and/or mean) and variability (interquartile range and/or mean absolute deviation), as well as describing any overall pattern and any striking deviations from the overall pattern with reference to the context in which the data were gathered.

General Procedures:

A.) Give students a *Whodunnit Worksheet*. Students will need to solve the 10 problems about the topic. Each correct answer will result in a clue. Students will check off the Players, Last Known Whereabouts and Methods that are accounted for. The one remaining from each column is the correct solution. In this case it should be:

“Professor Delta used chemical poisoning in the movie theater.”

B.) Print out the “Clues” and cut them out. I suggest you laminate them for future use. Spread them in different locations in the classroom and turn the students loose. Have teams of 2-3 circulate and attempt to figure out the clues. Once they think they have all the pieces, they can come to you to check their answer. If they are wrong, let them figure it out ☺

An alternative “Clue 10” is provided in the event you want a different result for different class periods.



Whodunnit?



Name : _____

Six contestants on a reality TV show were stunned to find their lowest scoring colleague was “injured.” They must figure out the crime before the bell rings. The question is *Whodunnit?* And how... The Player, Last Known Whereabouts and Method that are left unaccounted for -- is the solution.

1.	2.
3.	4.
5.	6.

7.

8.

9.

10.

Who is the "Criminal"?

(The three boxes left unchecked will reveal the crime. If you cannot figure it out, you may be next.)

The Players

Dr. Alpha	☐
Miss Beta	☐
Mrs. Gamma	☐
Professor Delta	☐
Mr. Epsilon	☐
Coach Omega	☐

**The Last Known
Whereabouts**

Atrium	☐
Gym	☐
Kitchen	☐
Library	☐
Movie Theater	☐
Pool	☐

The Method

Chemical Poisoning	☐
Cougar Attack	☐
Electrocution	☐
Fallen Object	☐
Stubbed Toe	☐
Venomous Bite	☐

???

Clue #1

W E O D U N N I I ?

Find the mode: 6,8,0,8,9,8

- | | |
|---|--|
| 0 | The injury was not caused by chemical poisoning. |
| 6 | Mr. Epsilon was reading in the library. |
| 8 | The victim was not injured by a cougar. |
| 9 | Coach Omega was exercising in the gym. |

W E O D U N N I I ?

Clue #2

Find the median: 1,2,4,5,6,7,8

- | | |
|---|--|
| 1 | The victim did not stub a toe. |
| 2 | The victim was not electrocuted. |
| 4 | Professor Delta was watching a movie in the theater. |
| 5 | Miss Beta was baking cookies in the kitchen. |

???

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Clue #3

W E O D U N N I I ?

Find the mean: 1,4,5,10

- | | |
|-----|--|
| 5 | The victim did not have a venomous bite. |
| 5.5 | Miss Beta was lounging around the atrium. |
| 6 | Mrs. Gamma was laying by the pool. |
| 10 | The injury was not because of a fallen object. |

W E O D U N N I I ?

Clue #4

Find the range: 0,1,4,9,11

- | | |
|----|--|
| 8 | Mr. Epsilon was eating a snack in the kitchen. |
| 9 | The victim did not have a venomous bite. |
| 11 | Dr. Alpha was taking notes in the library. |
| 12 | The victim was not attacked by a cougar. |

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Clue #5

W E O D U N N I I ?

Find the median: 9,10,5,12,6,5,3

- | | |
|----|--|
| 5 | Mr. Epsilon was reading by the pool. |
| 6 | The victim was not electrocuted. |
| 9 | Miss Beta was using a computer in the library. |
| 10 | The victim was not attacked by a cougar. |

W E O D U N N I I ?

Clue #6

Kirk scored 13, 14,16,15 and 19 in five of his math tests.
What is his mean score?

- | | |
|------|---|
| 14.6 | The injury was not caused by a fallen object. |
| 15.4 | Mrs. Gamma was playing the piano in the atrium. |
| 17 | The injury was not a stubbed toe. |
| 18 | Dr. Alpha was swimming in the pool. |

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Clue #7

W E O D U N N I I ?

A student got \$23, \$11, \$31, \$20 and \$17 on five consecutive days. What is the range of her pocket money?

\$20 The injury was not carried out with a fallen object.

\$21 The victim did not have a venomous bite.

\$22 Coach Omega was browsing books in the library.

\$23 Professor Delta was playing basketball in the gym.

W E O D U N N I I ?

Clue #8

Find the mode: 16,31,31,31,24,16,24,16,24,17,16

16 Mr. Epsilon was arm wrestling by the pool.

17 Miss Beta was reading in the library.

24 The injury was not a stubbed toe.

31 The victim did not have a venomous bite.

???

???

Clue #9

W E O D U N N I I ?

Which option has the same median and mode?

1,2,1,3,4 Coach Omega was watching a movie in the theater.

3,4,3,5,6 The injury was not done with chemicals.

6,6,7,9,8 Professor Delta was doing yoga in the gym.

9,9,9,2,2 The injury was not a stubbed toe.

W E O D U N N I I ?

Clue #10

Which option has the same mode and mean?

3,3,3 Coach Omega was shaking hands in the gym.

3,4,4 The injury was not done with chemical poisoning.

5,6,7,3,3 The victim did not stub a toe.

10,11,10 Professor Delta was playing a video game in the movie theater.

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Alternate Ending

W E O D U N N I I ?

"Coach Omega used chemical poisoning in the gym".

W E O D U N N I I ?

Clue #10

Which option has the same mode and mean?

3,3,3

Professor Delta was playing a video game in the movie theater.

3,4,4

The injury was not done with chemical poisoning.

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